Ex. no: 4
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SETUP AND CONFIGURE A LAN (Local Area Network) USING A SWITCH AND ETHERNET CABLES IN YOUR LAB

AIM:

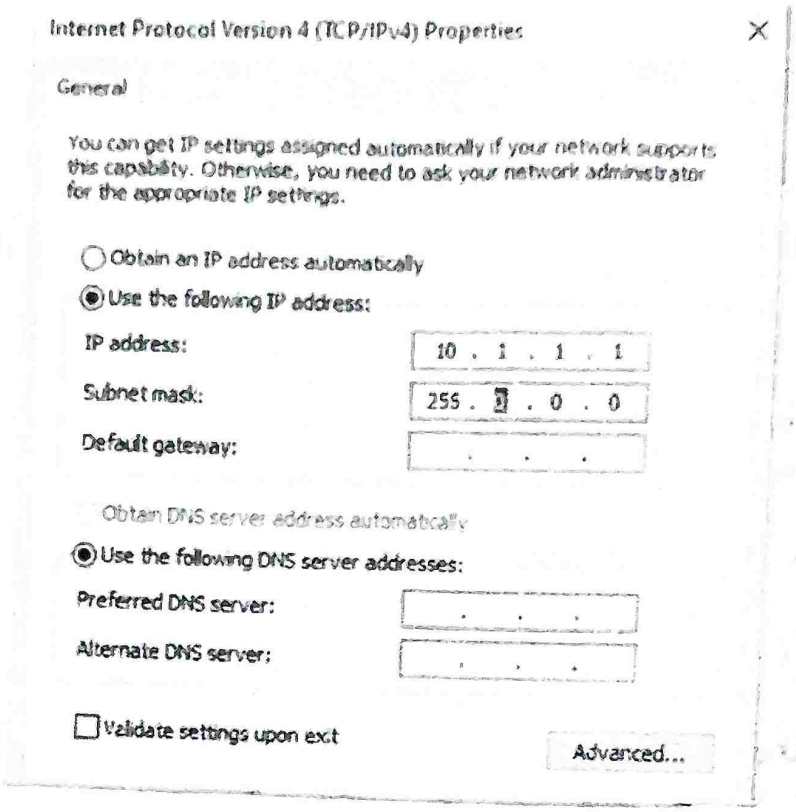
Setup and configure a LAN (Local Area Network) using a Switch and Ethernet Cables in your Lab.

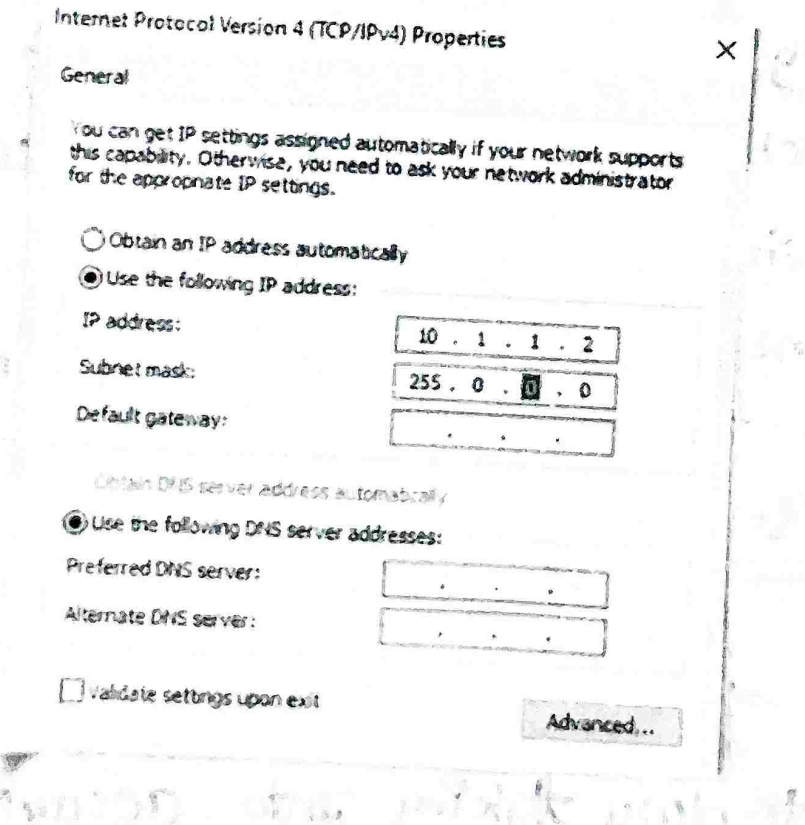
What is a LAN?

A Local Area Network (LAN) refers to a network that connects device within a limited area, such as an office building, school or home. It enables users to share resources, including data, printers and Internet access. A local area network (LAN) switch serves as the primary connecting device, managing and directing communications within the local network. Each connected device ona LAN switch can communicate directly with each other, allowing for fast & secure data transfer.

How to set up a LAN

Step 1: Plan and Design an appropriate network topology taking into account network requirements and equipment location.


Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address: 10 . 1 . 1 . 1
Subnet mask: 255 . 0 . 0 . 0
Default gateway:
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Validate settings upon exit
Advanced...


Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address: 10 . 1 . 1 . 2
Subnet mask: 255 . 0 . 0 . 0
Default gateway:
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Validate settings upon exit
Advanced...

Step 2 : You can take 4 Computers, Switch with 8, 16, or 24 ports which is sufficient for networks of those sizes, and 4 Ethernet Cables.

Step 3 : Connect your computers to the network switch via an Ethernet cable, which is as simple as plugging one end of the Ethernet cable into your computer and the other end into your network switch.

Step 4 : Assign IP address to your PCs.

1. Log on the client computer as Administrator or as Owner.
2. Click Network and Internet Connections
3. Right Click Local Area Connection / Ethernet → Go to Properties → Select Internet Protocol (TCP/IPv4) → Click on Properties → Select use the following ip address option and assign ip address.

Similarly assign IP address to all the PCs connected to switch.

eg:

PC1 - IP address : 10.1.1.1, Subnet mask 255.0.0.0.

Step 5 : Configure a network switch:

1. Connect your computer to the switch :
Connect your computer switch using an Ethernet cable.

2. Log in to the web interface : Open a web browser, and enter the IP address of the switch, Enter username, password and login.

3. Configure basic settings : On youre logged in, you will able to configure basic setting for the switch.

4. Assign IP address as : 10.1.1.5, Subnet mask 255.0.0.0.

Step 6 : Check the connectivity between switch and other machine by using ping command in the command prompt of the device.

Step 7 : Select a folder, → go to properties → click Sharing tab → share it with everyone on the same LAN.

Step 8 : Try to access the shared folder from others Computers of the network.

RESULT :

Thus, the setup and Configure a LAN. Using a switch and Ethernet cables has been Implemented and executed successfully.